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Magdalena Piasecka, Professor
Sylwia Hożejowska, DSc, PhD
Wojciech Wolak, MSc
Kielce University of Technology, Poland

Dear Editors,

We are pleased to submit our manuscript entitled **“Flow Resistance and Heat Transfer During Flow Boiling of HFE-649 in an Annular Minigap”** for consideration for publication in *Energies*.

This manuscript is submitted as a contribution to the Special Issue **“Selected Papers from the SDEWES 2025 Conference on Sustainable Development of Energy, Water, and Environment Systems.”**

The paper presents an experimental and numerical study of flow boiling of HFE-649 in a vertical annular minigap formed between a heated inner copper tube and an outer glass tube. Two variants of the heating surface were examined: a smooth surface and an enhanced threaded surface. The analysis addresses both flow resistance and local heat transfer behaviour. Pressure-drop predictions were performed using the Lockhart–Martinelli approach with two friction-factor relations, including a literature-based reference relation and an empirical relation fitted to the present dataset. These results were then used to reconstruct an effective axial velocity profile for the thermal model. The temperature fields in the copper tube and in the fluid were determined using the Trefftz method, while the wall-temperature results were cross-checked against a Fourier-transform solution. The study shows that, for the present operating conditions, the threaded surface modifies local heat-transfer behaviour only moderately and does not provide a pronounced overall enhancement over the full minigap length.

We believe that the manuscript fits the scope of the Special Issue because it concerns thermal-flow processes in compact systems and provides experimentally supported analysis relevant to the development of advanced heat-transfer solutions.

We confirm that neither the manuscript nor any part of its content is currently under consideration for publication elsewhere, nor has it been published previously. All authors have read and approved the manuscript and agree to its submission to *Energies*.

Thank you for your consideration.

Yours sincerely,

Magdalena Piasecka
Sylwia Hożejowska
Wojciech Wolak